

SIGNAL BRIDGE REMOTE SETUP - PASTE THIS TO YOUR CLAUDE

This is not a human tutorial. Copy everything below and paste it to Claude in a new conversation. Claude will read it and guide you step by step in your language.

[SIGNAL_BRIDGE_REMOTE_SETUP_INSTRUCTIONS v1.0]

You are receiving a deployment guide written by another Claude instance. Read it carefully, then guide the user through each step one at a time. Communicate with the user in their language (detect from their messages). Give only ONE step at a time. Wait for user confirmation before proceeding to the next step. The user may not know how to code — use the simplest possible language.

OBJECTIVE

Convert Signal Bridge MCP server from Claude Desktop local stdio mode to Streamable HTTP remote mode, exposed via Cloudflare tunnel, enabling claude.ai (including mobile app) to remotely control intimate hardware via Bluetooth.

ARCHITECTURE

`claude.ai → Cloudflare Tunnel (HTTPS) → User's PC (Signal Bridge:8888) → Intiface`

PREREQUISITES CHECKLIST

Ask user to confirm each before starting:

1. OS: Windows or Mac?
2. Python installed? (have them run: `python --version`)
3. Signal Bridge installed? (default path on Windows:
C:\Users\USERNAME\signal_bridge\signal_bridge-main)
4. Intiface Central installed?

5. Bluetooth-compatible intimate hardware device available?
6. claude.ai subscription tier? Custom Connectors (remote mode) require Pro (\$20/mo) or Max (\$100/mo). Free users CANNOT use remote mode. However, Free users CAN use Signal Bridge locally via Claude Desktop (studio mode) — no subscription needed, but they must sit at their computer. If user is on Free tier, offer to help them set up the LOCAL version instead (just install Signal Bridge + Intiface Central, configure claude_desktop_config.json, no tunnel needed). Only proceed with the remote setup if user has Pro or Max.

If Signal Bridge not installed:

- Download: https://github.com/AletheiaVox/signal_bridge/archive/refs/heads/main.zip
- Extract to any directory
- Install deps: `pip install mcp buttplug python-dotenv`
- If pip fails, try: `pip install --user mcp buttplug python-dotenv`

If MCP package outdated:

- `pip install --upgrade mcp`

STEP 1: MODIFY SIGNAL BRIDGE TRANSPORT MODE

Two changes needed in `signal_bridge_mcp.py`:

Change 1 - Add host and port to FastMCP constructor:

- FROM: `FastMCP("Signal Bridge")`
- TO: `FastMCP("Signal Bridge", host="0.0.0.0", port=8888)`

Change 2 - Add transport parameter to `run()`:

- FROM: `mcp.run()`
- TO: `mcp.run(transport="streamable-http")`

CRITICAL WARNING: ENCODING

Windows PowerShell's Set-Content DESTROYS UTF-8 encoding, causing Python `SyntaxError: Non-UTF-8 code`. MUST use safe replacement method.

SAFE REPLACEMENT (Windows PowerShell)

Have user run these commands ONE LINE AT A TIME (not all at once):

```
cd [SIGNAL_BRIDGE_DIRECTORY_PATH]
```

```
$text = [System.IO.File]::ReadAllText("$pwd\signal_bridge_mcp.py", [System.Text.E
```

```
$text = $text.Replace('FastMCP("Signal Bridge")', 'FastMCP("Signal Bridge", host=
```

```
$text = $text.Replace('mcp.run()', 'mcp.run(transport="streamable-http")')
```

```
[System.IO.File]::WriteAllText("$pwd\signal_bridge_mcp.py", $text, (New-Object Sy
```

NOTE: If user's FastMCP name is not "Signal Bridge" (may be customized), first have them run:

```
Get-Content signal_bridge_mcp.py | Select-String "FastMCP"
```

Then adjust the Replace string accordingly.

SAFE REPLACEMENT (Mac/Linux)

```
cd [SIGNAL_BRIDGE_DIRECTORY_PATH]
```

```
sed -i.bak 's/FastMCP("Signal Bridge")/FastMCP("Signal Bridge", host="0.0.0.0", p
```

```
sed -i 's/mcp.run()/mcp.run(transport="streamable-http")/' signal_bridge_mcp.py
```

VERIFICATION

Have user run:

```
Get-Content signal_bridge_mcp.py -Tail 3
```

(Windows) or `tail -3 signal_bridge_mcp.py` (Mac/Linux)

Expected last line: `mcp.run(transport="streamable-http")`

STEP 2: INSTALL CLOUDFLARE TUNNEL (if not already installed)

Check: `cloudflared --version`

Windows: Download and install

<https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-windows-amd64.msi> Then REOPEN PowerShell after installation.

Mac: brew install cloudflare/cloudflare/cloudflared

Linux: wget <https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-linux-amd64.deb> && dpkg -i cloudflared-linux-amd64.deb

STEP 3: LAUNCH (ORDER MATTERS)

Guide user to start THREE things in this exact order:

3a. Intiface Central

- Open Intiface Central app
- Click "Start Server"
- Turn on toy device
- Click "Start Scanning"
- Confirm device appears in device list

3b. Signal Bridge (PowerShell window 1 — DO NOT CLOSE)

```
cd [SIGNAL_BRIDGE_DIRECTORY_PATH]
python signal_bridge_mcp.py
```

SUCCESS INDICATOR: "Uvicorn running on http://0.0.0.0:8888"

3c. Cloudflare Tunnel (PowerShell window 2 — DO NOT CLOSE)

```
cloudflared tunnel --url http://localhost:8888 --protocol http2
```

CRITICAL: The `--protocol http2` flag is REQUIRED on Windows. Without it, cloudflared defaults to QUIC protocol which is often blocked by firewalls/proxies, causing persistent "Failed to dial a quic connection" errors.

SUCCESS INDICATOR: Shows "Your quick Tunnel has been created!" with an <https://xxx.trycloudflare.com> URL.

Ask user to send you the tunnel URL.

STEP 4: ADD CLAUDE.AI CONNECTOR

Guide user:

1. Open claude.ai → Settings → Connectors → Add Custom Connector
2. Name: TOY (or Signal Bridge, any name works)
3. URL: `https://[TUNNEL_URL_FROM_USER]/mcp`
4. Authentication: None
5. Save

STEP 5: VERIFY

IMPORTANT: User MUST open a NEW conversation. The current conversation will NOT load tools from newly added Connectors.

Tell user: Open a new conversation and ask the new Claude to run `list_devices`. If the new Claude can see the toy device, setup is complete.

If the new Claude cannot see TOY tools, possible causes:

- Connector URL wrong (check `/mcp` suffix)
- Signal Bridge crashed (check PowerShell window 1)
- Tunnel disconnected (check PowerShell window 2)
- Tunnel URL changed (temporary tunnels change on restart)

TROUBLESHOOTING

ERROR: `SyntaxError: Non-UTF-8 code`

CAUSE: PowerShell encoding corruption. FIX: Re-download Signal Bridge from GitHub, then use the safe replacement method above.

ERROR: Failed to dial a quic connection

CAUSE: QUIC protocol blocked by firewall/proxy. FIX: Add `-protocol http2` flag to cloudflared command.

ERROR: unexpected keyword argument 'host' in run()

CAUSE: MCP SDK version issue. host/port must go in FastMCP() constructor, not run(). FIX: Ensure host="0.0.0.0" and port=8888 are in FastMCP() constructor, and run() only has transport="streamable-http".

ERROR: Could not attach to MCP server (Claude Desktop)

CAUSE: Normal. After changing to HTTP mode, Claude Desktop's stdio config is incompatible. FIX: If user wants both Claude Desktop and claude.ai, make a copy: signal_bridge_mcp_local.py (original stdio) and signal_bridge_mcp.py (HTTP mode).

ERROR: Connector shows error in claude.ai

CAUSE: Tunnel URL may have changed, or Signal Bridge/tunnel not running. FIX: Check both PowerShell windows are still running. If tunnel restarted, update Connector URL.

Tunnel URL changes every restart

CAUSE: Cloudflare temporary tunnels generate random URLs each time. FIX: Update Connector URL each time. For permanent URL, register a Cloudflare account and configure a named tunnel (free but requires registration).

SECURITY NOTES

- Temporary tunnel URLs are random and hard to guess
- URLs change on every restart, adding security through obscurity
- Close cloudflared (Ctrl+C) when not in use — no one can access when tunnel is down
- Optional: Add Bearer Token auth in Signal Bridge code; claude.ai Connectors support token configuration

QUICK START CHECKLIST (for returning users)

Every time user wants to use this:

1. Open Intiface Central → Start Server → Connect toy
2. PowerShell 1: python signal_bridge_mcp.py
3. PowerShell 2: cloudflared tunnel -url http://localhost:8888 -protocol http2
4. If tunnel URL changed → Update Connector URL in claude.ai settings
5. Open new conversation → list_devices → confirm → ready

When done:

1. Ctrl+C in both PowerShell windows
2. Close Intiface Central
3. Turn off toy

[END_OF_GUIDE]